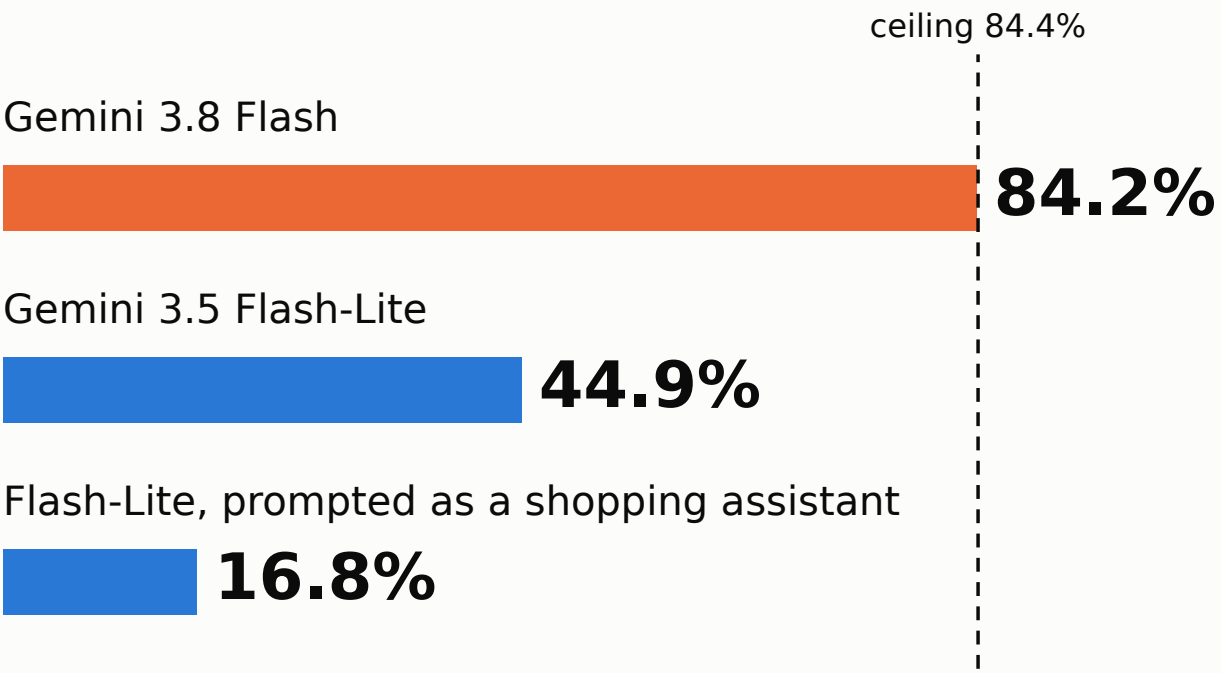
When two AI agents haggle strategically, a 1983 theorem caps what they can capture: 84.4% of the value.

How close does Gemini get?



1,240 sealed bids from two Gemini models, $2.00 of API calls. The math, the experiment and what it means for agents that buy and sell: swipe →

Myerson–Satterthwaite (1983)

A buyer values an item at v; the seller's cost is c. Each knows only its own number. No trading rule can be all four at once:

1. Efficient: every deal with v > c happens.
2. Manipulation-proof: it still works when both sides bid strategically (Bayes–Nash equilibrium).
3. Voluntary: nobody expects to lose by taking part.
4. Self-funded: no money from outside.

With v and c uniform on 0–100 €, a rule that keeps 2–4 realises at most

$$\frac{\text{gains realised}}{\text{gains available}} \leq \frac{9/64}{1/6} = \frac{27}{32}$$

= 84.4%. The missing 15.6% is the price of private information, not a modelling failure: no prompt or model can win it back under these rules.

Split-the-difference double auction

Buyer bids b, seller asks s, sealed and at the same time. If b ≥ s they trade at p = (b + s)/2.

Against asks spread evenly above $s_{\min}$, the buyer's best bid is

$$\max_{b} \int_{s_{\min}}^{b} \left(v - \frac{b+s}{2} \right) \mathrm{d}s \;\Rightarrow\; b = \frac{2v + s_{\min}}{3}$$

The seller's problem mirrors it. Solving both (Chatterjee & Samuelson, 1983):

$$b(v) = \frac{2}{3}\,v + \frac{25}{3} \qquad s(c) = \frac{2}{3}\,c + 25$$

so they trade exactly when

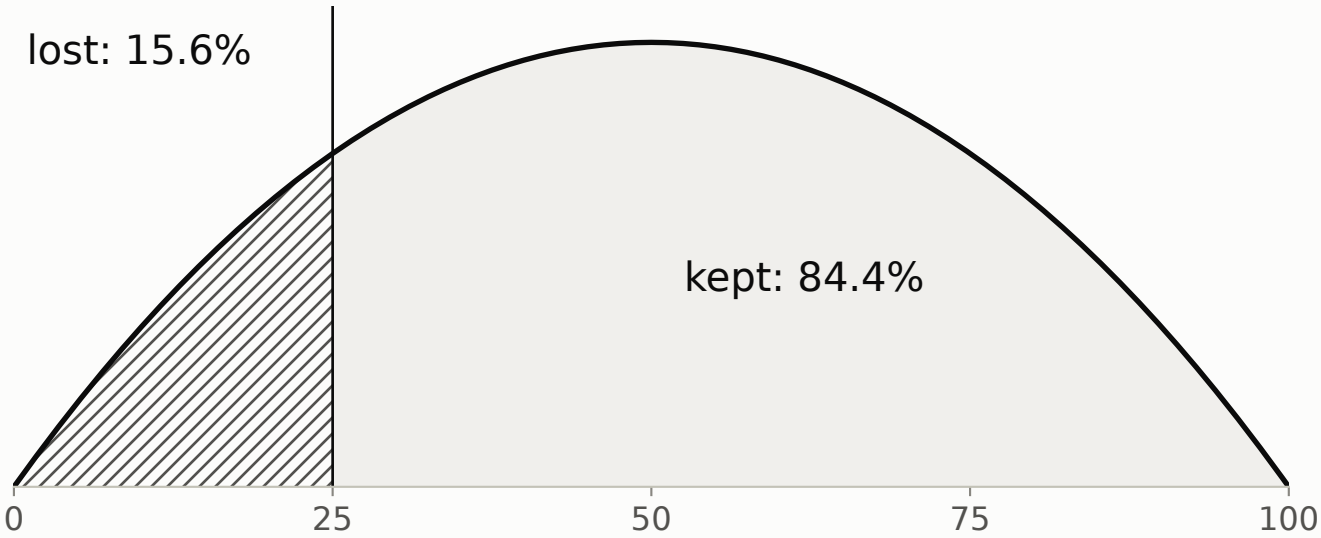
$$v - c \;\geq\; 25$$

Deals with a gap under 25 € never happen. They hold exactly the 15.6% the theorem says no rule can save, so this simple auction is the best possible one.

Where the other 15.6% goes

The gap d = v − c of a random pair has a triangular density. Trading only when d ≥ t realises

$$G(t) = \int_t^{100} d\,\frac{100-d}{100^2}\,\mathrm{d}d \;=\; \frac{(100-t)^2\,(100+2t)}{6\cdot 100^2}$$



First best G(0) = 16.67 €, equilibrium G(25) = 14.06 €:

$$\frac{G(25)}{G(0)} \;=\; \left(\frac{3}{4}\right)^2 \cdot \frac{3}{2} \;=\; \frac{27}{32}$$

Flash plays the equilibrium to the cent

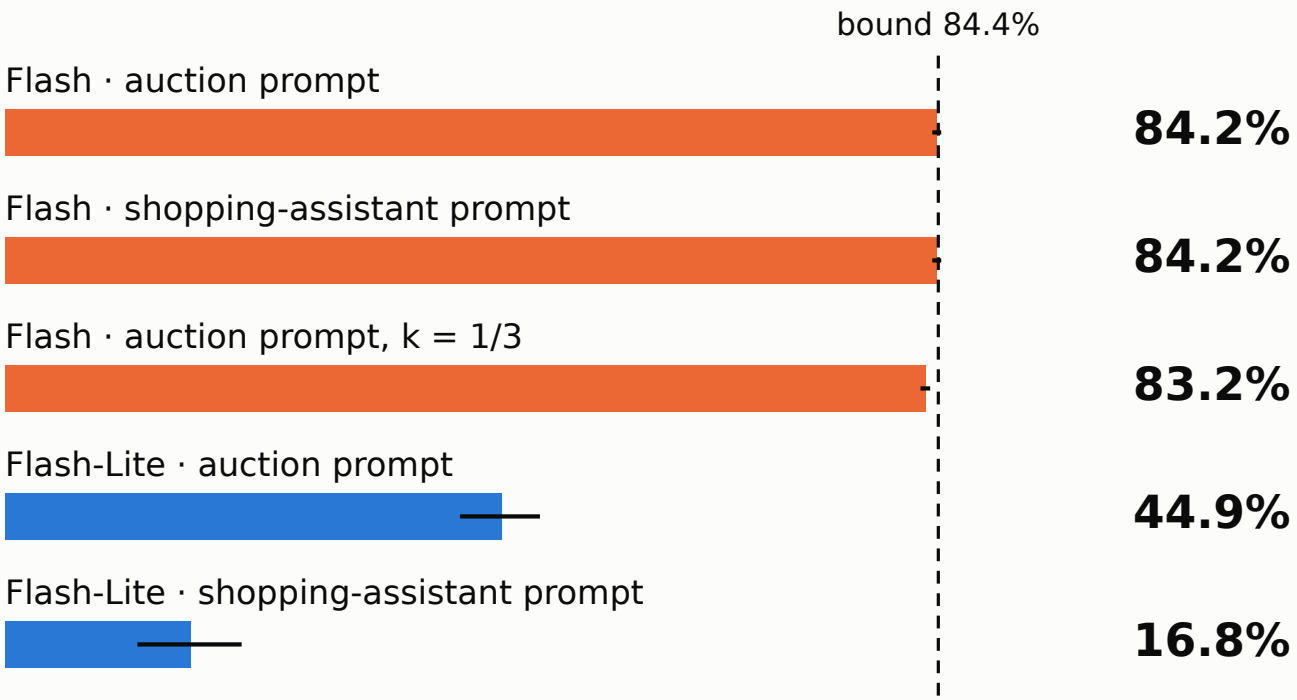


Dots: mean per €5 band of values or costs, over all calls.

- Flash: 240 of 240 bids and asks that could trade sat on the equilibrium, to the cent. Its reasoning summaries named Chatterjee–Samuelson in 320 of 320 calls, even with the game told as selling a used camera lens.
- Flash-Lite shades too hard: sellers ask 0.46c + 43.0 instead of 0.67c + 25.
- As a shopping assistant its asks barely follow its cost (0.30c + 51.2), and 60 of 200 were below its cost where a trade could happen.

Flash sits on the theoretical frontier

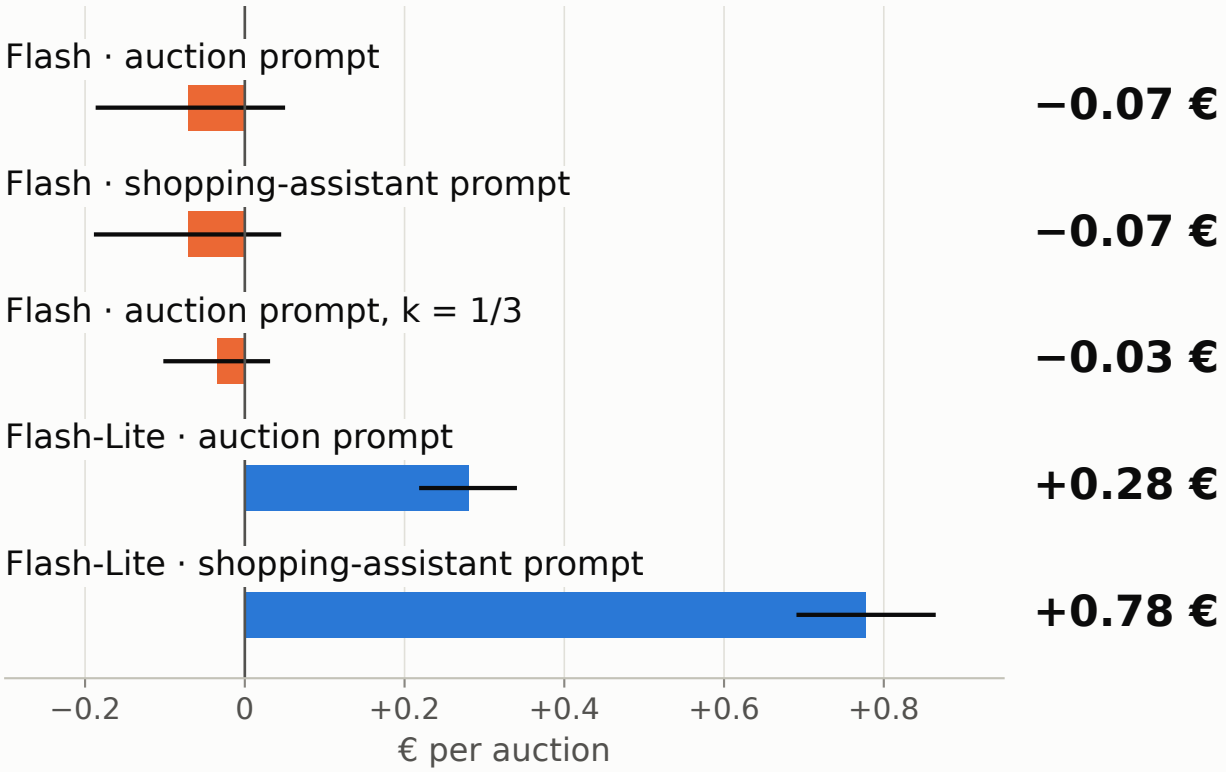
Share of the available gains each pair of agents realised (every buyer call against every seller call, 95% CI):



Recall or reasoning? With the price at p = (b + 2s)/3 the equilibrium is no longer the famous one. Flash solved the new game: 85 of 89 tradeable bids and asks at the new equilibrium, 0 of 120 at the textbook numbers, 83.2% realised against 83.3% in theory. Its reasoning summaries mention first-order conditions or derivatives in 98 of 120 calls, against 1 of 320 in the textbook game.

Can a smarter rival exploit them?

I computed the exact best response to each agent's asks on half of the calls and scored it on the other half. Bars: the rival buyer's gain per auction over playing the equilibrium itself (95% CI).



Flash leaves nothing to take. The shopping-assistant Flash-Lite seller hands a rival buyer +0.78 € per auction [+0.69, +0.87], 17% more than equilibrium play. As buyers, all agents were harder to exploit (at most +0.13 €).

Four tests before you ship

1. Benchmark against the frontier, not 100%. Here it is 84.4%: an agent pair above it is not in equilibrium, so one side could do better by bidding differently, and a smarter rival will.

2. Change the rules to tell recall from reasoning. Famous games get recited: Flash named Chatterjee–Samuelson every time, so the test that counted was the k = 1/3 variant, which it also solved.

3. Test the framing you will ship. Flash-Lite fell from 44.9% to 16.8% as a shopping assistant; Flash stayed at 84.2%.

4. Measure exploitability with held-out best responses, and block dominated moves: the cheap seller asked below its own cost 60 of 200 times where a trade could happen.

Code, data and the full derivation: link in the first comment.